Aaryan Muku

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Education

University of Alberta — Edmonton, Canada

BSc Computing Science Specialization | GPA: 3.5/4.0, last 30 credits — *Sep 2023 – May 2028*

- Coursework: Algorithms, Machine Learning, Reinforcement Learning, Full-Stack Development, Database Management, AI Planning, Information Retrieval, Distributed Systems, and Systems Programming.

Experience

Software Developer, Cloud Development — May 2026 – Dec 2026

Bank of Montreal (BMO) — *Toronto, Canada*

- Built ECI complete migration components moving legacy dynamodb banking data into multi-account dynamodb AWS structures for more than 30 Million records with validation, retry, reconciliation, error handling and release safeguards.
- Designed Infrastructure flows using SQS, SNS, Lambda, S3, DynamoDB, ECS, and data-sync components to support BMO's new multi-account ECI platform.
- Automated record-level validation and reconciliation checks, reducing manual migration review by approximately 40% across testing cycles and release reviews.
- Improved Lumi, BMO's AI assistant, by refining cloud support flows, platform knowledge retrieval and improved response time.
- Implemented business procedure components for AML compliance, improving traceability across account, customer, and exception-management workflows.
- Led ad hoc production deployments for active-consumer platform components, owning post-production validation, release-health monitoring, issue triage, and rollback-readiness checks.
- Introduced data-integrity checks and failure-handling paths for migration, sync, and recon services to improve audit readiness, observability, and reliability.

Software Developer Intern — May 2025 – Aug 2025

ARO Robotic Systems — *Edmonton, Canada*

- Integrated the Omni-5 robotic arm into the robot stack using ROS2 nodes, C++/Python control services, and perception-to-actuation messaging for hardware.
- Built ROS2 architecture for arm control, telemetry, state management, feedback, and low-latency communication between perception and planning modules.
- Developed movement logic with trajectory generation and collision-aware planning, improving manipulation reliability by approximately 25% in repeated tests.
- Developed C++/Python motion services connecting perception, planning, and actuation layers, reducing integration testing by over 30% during validation.
- Implemented structured logging, health checks, and fault-recovery workflows across sensor, controller, and motion-planning services for faster debugging.
- Collaborated with controls, robotics, and hardware teams to tune Omni-5 movement, validate safety constraints, and stabilize full robot execution.
- Helped improve movement behavior through repeated hardware trials, ML feature tuning, and feedback from perception/control loops during arm integration.

Machine Learning Intern — May 2024 – Aug 2024

Naukri.com, InfoEdge India Ltd. — *Delhi NCR, India*

- Built anomaly detection for Data News, surfacing high and low traffic spikes for jobseeker metrics across locations, segments, and time periods at platform scale.
- Engineered Spark and Python pipelines transforming sessions, visitors, jobseeker activity, holidays, and seasonality into forecasting features for Prophet.
- Implemented Facebook Prophet with trend, seasonality, holiday, and location-aware regressors, improving spike detection to approximately 95% accuracy.
- Feature-engineered anomaly logic to separate recurring demand cycles from real spikes, reducing noisy percentile-threshold alerts for analytics investigation.
- Packaged spike outputs into Data News monitoring workflows so teams could inspect unusual user traffic patterns across regions faster and with context.
- Validated model behavior with analytics stakeholders across national traffic patterns, improving alert quality, investigation readiness, and stakeholder trust.
- Benchmarked Prophet with base anomaly methods and tuned thresholds for week, month, year seasonal, holiday, and region-specific traffic patterns in prod data.

Software Development Intern — May 2022 – Aug 2022

Kent RO Systems Ltd. — *Delhi NCR, India*

- Developed Java features and bug fixes for Cameye, Kent's home-camera app, improving device controls, dashboard reliability, and user workflows across releases.
- Introduced 360-degree camera rotation across the mobile app and dashboard, connecting UI actions with camera movement, device state, and control feedback.
- Resolved monitoring, control, notification, and device-operation issues, improving reliability across connected home-camera usage flows and customer scenarios.
- Built IoT dashboard components for live device state, camera operations, customer monitoring, and cloud-to-device communication workflows for operators.
- Collaborated with app, backend, and embedded teams to debug production issues across mobile, dashboard, cloud, and hardware-facing layers.
- Improved dashboard interaction flows for home-camera operators by tightening device-state feedback, camera-control response behavior, and UI reliability.
- Tested camera-control fixes across device states and network conditions to reduce regressions in monitoring, rotation, and control flows before release.

Projects

BlackBox — *Kubernetes, AWS, Databases, Fault Injection, Observability*

- Building local Kubernetes disaster tournaments that replay outages and score rollback, restore, scaling, and database-recovery strategies side by side.
- Designing workload replay, snapshots, fault injection, traces, lost-write checks, recovery-time scoring, correctness, and cloud-cost analysis dashboards.
- Developing isolated incident branches for commerce workloads with PostgreSQL, queues, observability, and recovery-comparison dashboards for engineers.
- Comparing recovery plans against business invariants so availability fixes do not introduce duplicate writes, lost payments, or unsafe retries.
- Modeling failures that look like compute shortages but actually involve database contention, queue retries, and unsafe payment behavior under load.

Ripple — *LLMs, MCP, RAG, Git, Code Intelligence*

- Building MCP/RAG code intelligence that maps repos, schemas, APIs, and architecture flows before agents plan connected code changes across services.
- Designing Git-aware agents for repo inspection, database awareness, dependency tracing, impact analysis, and patch propagation through pull requests.
- Developing local-first workflows for codebase search, architectural reasoning, change plans, test selection, and pull-request preparation from one machine.
- Adding MCP tools for Git, database inspection, code search, test awareness, and change-risk scoring across connected components and dependencies.
- Building architecture graphs that connect source files, APIs, migrations, database tables, tests, owners, and deployment dependencies for change safety.

Packet — *Python, FastAPI, RAG, LangChain, Multi-Agent Systems*

- Building a multi-agent intelligence layer that connects scattered documents and assembles unified, context-rich knowledge packets.
- Designing specialized agents to reconcile fragmented information through semantic retrieval, entity linking, and cross-source synthesis.
- Developing orchestration workflows that split research requests into tasks, coordinate agents, and merge findings into reusable knowledge packets.
- Integrating RAG pipelines with semantic search, embeddings, and provenance tracking to ground each packet in verifiable source documents.
- Implementing conflict detection and evidence reconciliation to surface duplicate claims, contradictions, missing context, and source reliability.

Crimson — *Django, REST APIs, PostgreSQL, Distributed Systems*

- Built a federated social platform allowing independently operated nodes to exchange posts, comments, likes, follows, and privacy controls across servers.
- Implemented Django REST services with push-based federation, node authentication, follow approvals, and cross-node synchronization for distributed content.
- Designed normalized, indexed PostgreSQL schemas supporting distributed consistency, query performance, and administrative workflows for node operators.
- Developed moderation and administration flows for node access, content visibility, deleted-entry privacy, and federated relationship management.
- Built frontend and backend workflows for author identity, private posts, remote content visibility, node approvals, and deleted-entry handling.

Technical Skills

Languages: Python, C, C++, Java, JavaScript, TypeScript, SQL, Bash, R, HTML/CSS
Backend/Frameworks: FastAPI, Flask, Django, Node.js, Express, React, Next.js, Tailwind CSS, Streamlit, ROS2, REST APIs
ML/Libraries: PyTorch, scikit-learn, XGBoost, LightGBM, Transformers, LangChain, Prophet, FastF1, NumPy, Pandas, Matplotlib
Data/Databases: Apache Spark, PostgreSQL, MongoDB, MySQL, Redis, Prisma, Firebase
Cloud/DevOps: AWS, Azure, GCP, Docker, Kubernetes, GitHub Actions, Git, Linux, CI/CD, Vercel, Netlify
AI/Agents/Systems: MCP, RAG, OpenTelemetry, distributed tracing, fault injection, workload replay, multi-agent orchestration